SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO2: *Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)*

Assessment Tool Weightage: 50%

Duration: 1 Hour

QUESTIONS: 5 **Total Marks:50**

Annexure A

Scenario Based Assignment

Code of Conduct:

I Mohammad Aleyas (192521220) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Mohammad Aleyas

Annexure A

Scenario Based Assignment

1. Scenario : A government deploys AI-powered drones to deliver emergency medicines in a flood-affected region. The environment is highly dynamic—roads may suddenly become inaccessible, and weather conditions affect travel cost. The drone has access to: A heuristic estimate (straight-line distance), Partial real-time updates about blocked paths and Variable movement costs depending on terrain and weather. However, due to urgency, the drone must quickly decide routes with minimum delay and risk.

Tasks:

i. Explain how Greedy Best-First Search would behave in this scenario and discuss its strengths and weaknesses under uncertainty

ii. Explain how A* would handle the same situation, including how it balances cost and heuristic

iii. Analyze the impact of heuristic accuracy on both algorithms

iv. Discuss how dynamic changes (blocked paths) affect search performance

v. Recommend the most suitable algorithm and justify your decision considering real-time constraints, optimality, and safety

2. **Scenario**: A smart city implements an AI system to optimize traffic signal timings across hundreds of intersections. The objective is to minimize Total waiting time, Fuel consumption and Traffic congestion. The system must continuously adjust signals based on traffic flow. However the search space is extremely large, traffic patterns change dynamically and the system may get stuck in locally optimal solutions

Tasks. Formulate this as an optimization problem and explain why traditional search is inefficient

i. Explain how Hill Climbing would work in this scenario and identify its limitations

ii. Discuss issues like local maxima, plateaus, and ridges with real-world interpretation

iii. Explain how Simulated Annealing or Genetic Algorithms can overcome these limitations

iv. Recommend the best approach and justify your choice based on scalability and adaptability

3. **Scenario:** An autonomous rover is deployed on Mars to explore unknown terrain. It must: navigate safely to collect samples, avoid hazards (craters, rocks), operate with limited sensing capability and make decisions without a complete map. The rover updates its knowledge as it explores, making this a real-time decision-making problem.

Tasks:

i. Explain why this problem requires an online search agent instead of offline search

ii. Analyze the challenges of partial observability and dynamic environments

iii. Describe how the rover builds and updates its internal model

iv. Compare online search with uninformed and informed search strategies

v. Justify the most suitable approach considering uncertainty, adaptability, and safety

4. **Scenario:** A university is designing an AI system to automatically generate exam timetables. The system must satisfy multiple constraints: No student should have overlapping exams, Limited number of exam halls, Certain subjects must be scheduled before others, Faculty availability constraints, Special accommodations for some students, The complexity increases as the number of courses and students grows.

Tasks :

Formulate the problem as a Constraint Satisfaction Problem (CSP)

i. Clearly define variables, domains, and constraints with explanation

ii. Explain how backtracking search works in this scenario

iii. Discuss how constraint propagation (e.g., forward checking) improves efficiency

iv. Analyze real-world challenges and justify the best strategy for scalability

5. **Scenario:** Strategic Game AI for Real-Time Decision Making (Minimax & Alpha-Beta Pruning) A gaming company is developing an AI opponent for a real-time strategy game. The AI must complete against human players, Make decisions within strict time limits, Evaluate complex game states with multiple possible moves, The decision tree is extremely large, making computation expensive.

Tasks:

i. Explain how the Minimax algorithm models this problem

ii. Analyze the computational challenges of large game trees

iii. Explain how Alpha-Beta pruning improves efficiency with detailed reasoning

iv. Design an evaluation function and justify the factors considered

v. Recommend a strategy for balancing optimality and real-time performance

ANSWERS

1. Emergency Medicine Drone Route Planning

i. Greedy Best-First Search selects the node with the smallest heuristic h(n), such as straight-line distance to the hospital. It can be fast and useful when time is critical, but it ignores the cost already travelled and may choose a short-looking route that is blocked or dangerous. It is not guaranteed to find the lowest-cost route.
ii. A* evaluates f(n) = g(n) + h(n), where g(n) is the cost already travelled and h(n) estimates the remaining cost. It balances known travel cost with estimated distance. With non-negative costs and an admissible/consistent heuristic, it can find an optimal route in a static problem.
iii. A more accurate heuristic guides the search toward the goal and reduces unnecessary exploration. An overestimated heuristic can cause A* to lose its optimality guarantee. Greedy search is strongly affected by misleading estimates because it relies only on h(n).
iv. When a path becomes blocked, the route must be updated and affected nodes reconsidered. Repeated replanning increases computation, so the drone should use current sensor data and a real-time replanning method.
v. A* with frequent replanning is the best general choice because it considers both route cost and remaining distance. Safety constraints should reject blocked or high-risk routes, and a time-bounded fallback should be available if immediate action is required.

2. Smart-City Traffic Signal Optimization

Formulation: a state is the current timing plan and traffic conditions; actions change green-light durations or signal phases; the objective function minimizes waiting time, fuel use, and congestion subject to safety and timing constraints. Traditional exhaustive search is inefficient because the number of intersections and possible timings creates a huge search space that changes over time.
i. Hill Climbing starts with a timing plan and repeatedly moves to a neighboring plan that improves the objective. It is simple and fast, but may stop at a local optimum and does not guarantee the global best plan.
ii. A local maximum is a timing plan better than nearby plans but worse than a distant plan. A plateau is a region where neighboring plans have similar scores, so progress stalls. A ridge is a narrow path of improvement that may require coordinated changes across several signals.
iii. Simulated Annealing sometimes accepts a worse move early on to escape local optima, gradually reducing that chance. Genetic Algorithms maintain a population of timing plans and use selection, crossover, and mutation to explore alternatives.
iv. A practical recommendation is a Genetic Algorithm or hybrid approach for offline/periodic optimization, combined with a fast local adjustment method using live traffic data. This balances broad exploration with quick adaptation.

3. Mars Rover Online Search

i. Offline search assumes the map and action outcomes are known in advance. The rover does not have a complete map, so it must act, observe, update its knowledge, and plan again. This is online search.
ii. Partial observability means sensors reveal only part of the terrain; readings may be noisy and hazards may be hidden. The rover must handle uncertain movement, limited power, communication delay, and the risk of entering unsafe areas.
iii. The rover stores a map or internal model of visited locations, obstacles, estimated travel costs, and unexplored areas. After each movement it processes sensor readings, updates the map, and replans from its new position.
iv. Uninformed search uses no estimate of distance to the goal and may explore many irrelevant states. Informed search uses a heuristic to prioritize promising locations, but usually assumes more knowledge. Online search combines planning with action and learning in the unknown environment.
v. A cautious online agent using heuristic guidance and continual replanning is suitable. It should prioritize safe reachable areas, update its model after every observation, and keep a safe fallback route when uncertainty is high.

4. University Exam Timetable as a CSP

i. Variables: the exam time slot and room for each subject. Domains: available time slots and suitable rooms. Constraints: no student has overlapping exams; room capacity is sufficient; rooms and faculty are available; prerequisite subjects are ordered correctly; and accommodations are respected.
ii. Backtracking assigns a time and room to one exam, checks constraints, and proceeds to the next. If an assignment creates a conflict, it undoes that assignment and tries another value.
iii. Forward checking removes values from the domains of unassigned exams that conflict with the latest assignment. If any domain becomes empty, the algorithm detects failure early and backtracks.

iv. Large numbers of subjects, changing availability, and many student conflicts make scheduling difficult. Use the most-constrained-variable heuristic, least-constraining-value ordering, forward checking, and conflict-aware optimization to improve scalability.

5. Strategic Game AI: Minimax and Alpha-Beta Pruning

i. Minimax models two opponents with opposing goals. The AI selects moves that maximize its score, assuming the opponent chooses moves that minimize it. Scores are propagated upward from terminal or depth-limited states.

ii. With branching factor b and search depth d, a basic Minimax search examines O(b^d) nodes. Large trees can exceed the available time and memory, especially in real-time games.

iii. Alpha-Beta pruning tracks alpha, the best value found for the maximizing player, and beta, the best value for the minimizing player. When alpha is at least beta, the remaining branches cannot change the decision and can be skipped. With good move ordering, it examines far fewer nodes while returning the same result as Minimax at the searched depth.

iv. A sample evaluation function is: score = 3 × resource advantage + 5 × unit advantage + 4 × territory advantage + 6 × objective progress − 5 × immediate threat. Weights can be tuned through testing; higher values represent strategically more important factors.

v. Use Alpha-Beta pruning with good move ordering, a depth limit, iterative deepening, and a heuristic evaluation function. Iterative deepening provides the best completed move if time expires, helping balance decision quality and response speed.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand